

Breadth Momentum and Hidden Markov Regime Detection: A Composite Signal for Tactical Equity Allocation

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Abstract

We apply a composite regime gate to a tactical allocation strategy on a universe of 19 country equity exchange-traded funds (ETFs). The composite is constructed by combining two regime detectors, the breadth-momentum signal and a Hidden Markov Model (HMM). We show that the two signals are complementary rather than redundant. Breadth is a cross-sectional measure of how broadly the universe is participating, while HMM is a time-series measure of the statistical state of the basket. Each signal alone produces many false positive bear flags, while the combination of two signals agrees at the major regime transitions of the backtested period. We use 4 equity-based features and 2 rate-based features. The defensive leg rotates between long-duration Treasuries (TLT) and cash. Over the period from January 2004 to December 2025, the composite strategy delivers a Sharpe ratio of 0.72 and a maximum drawdown of -23.1% , compared to 0.53 and -58.0% for the equal-weighted basket, 0.46 and -57.4% for MSCI EAFE, and 0.77 and -50.8% for SPY buy-and-hold. The composite strategy more than halves the maximum drawdown of the passive benchmarks and exceeds SPY's compound annual growth rate by 1.5%.

Keywords: tactical asset allocation, regime detection, hidden Markov model, breadth momentum, country momentum, trend following

JEL Classification: G11, G12, G17, C58

1. Introduction

Tactical asset allocation (TAA) is a dynamic investment strategy that actively adjusts a portfolio's asset allocation in response to changing market conditions. TAA aims to achieve higher risk-adjusted returns than passive investment approaches by systematically adjusting exposure.

Practitioner approaches to TAA use cross-sectional characteristics of the asset universe as regime indicators. Faber (2007) showed that simple price-based timing rules applied to a multi-asset universe can deliver equity-like returns with reduced drawdowns. Antonacci (2014) introduced the dual-momentum framework, which combines absolute and relative momentum signals. Keller and Keuning (2017, 2018) proposed breadth momentum that counts the number of universe constituents exhibiting positive momentum as an alternative to individual asset trend signals. Strategies such as Vigilant Asset Allocation (VAA), Defensive Asset Allocation (DAA), and their subsequent variants are widely used.

Academic research typically employs time-series properties of the broad market to infer latent regime states. Hamilton (1989) introduced regime-switching models for macroeconomic time series. Ang and Bekaert (2002) applied similar methods to international asset allocation. Guidolin and Timmermann (2007) showed that regime-switching models enhance allocation decisions across equity, bond, and T-bill universes. More recent work by Bulla et al. (2011) and Nystrup et al. (2015, 2017, 2018) have refined Hidden Markov Model (HMM) specifications for tactical allocation, including techniques such as walk-forward training, persistent-state penalization, and feature-saliency selection.

There is limited overlap between practitioner approaches and academic literature. Practitioner TAA papers in the breadth-momentum literature do not employ HMMs, while academic regime-

switching research do not use breadth signals. A small number of recent papers by Kim et al. (2019), Wang et al. (2020), and Gupta et al. (2025) combine HMM regime detection with active allocation rules. Kim et al. apply HMMs to time individual assets in a multi-asset allocation. Wang et al. use HMMs to switch between factor investing models on the S&P 500. Gupta et al. combine HMMs with tree-based ensemble classifiers through a voting framework on a single US index, using a feature set that includes a breadth-flavored input (TRIN).

We believe that breadth-momentum and HMM regime signals provide complementary information. Breadth-momentum is a cross-sectional signal. It measures how many assets in the universe are in positive trend states and tells us how healthy the universe is. In contrast, the HMM is a time-series signal. It measures the latent statistical state of the basket, and tells us what regime the market is in. It is possible for the asset universe to exhibit low breadth while the HMM regime remains bullish. Combining both signals is a good way to detect regime transitions that may be overlooked by relying on a single signal.

We test this hypothesis on a universe of 19 country equity ETFs and a defensive leg that rotates between long-duration Treasuries (TLT) and cash. When both regime signals indicate a bullish state, the composite strategy holds an equal-weighted sleeve of the three highest-momentum equity ETFs. When both signals indicate risk-off, the composite strategy rotates to the defensive asset. We compare the composite strategy's performance against passive benchmarks (SPY, MSCI EAFE, and the equal-weighted basket of the 19 country ETFs). We also compare the composite strategy's performance against breadth-only and HMM-only strategies, and decompose the contribution of each signal across regime transitions. In addition, we perform robustness checks across the parameter choices.

The remainder of the paper is structured as follows. Section 2 reviews the literature on breadth-momentum and HMM-based regime detection. Section 3 describes the data and the asset universe. Section 4 outlines the methodology, including the breadth signal, HMM specifications, and composite regime gate. Section 5 presents the main empirical results, including a decomposition of the composite signal’s contribution. Section 6 reports robustness checks. Section 7 discusses the limitations of the current work and potential extensions. Section 8 provides a conclusion for our work.

2. Related Literature

In this section we will review the relevant literature. The academic foundations of momentum are reviewed in Section 2.1. The practitioner breadth-momentum approach that motivates the breadth signal is reviewed in Section 2.2. The HMM regime detection literature, which provides the time-series regime signal, is reviewed in Section 2.3. The country equity momentum literature, which justifies the country equity ETF universe employed here, is reviewed in Section 2.4.

2.1 Academic foundations of momentum and trend-following

Cross-sectional momentum was first demonstrated by Jegadeesh and Titman (1993), who found that past winners outperform past losers over 3- to 12-month horizons. Asness, Moskowitz, and Pedersen (2013) expanded these findings to multiple asset classes, including country indices, bonds, currencies, and commodities.

The time-series momentum literature, originating with Moskowitz, Ooi, and Pedersen (2012) and extended back to 1880 by Hurst, Ooi, and Pedersen (2017), shows that an asset's own historical return can predict its future return. Babu et al. (2020) provide robust out-of-sample replication of

these results. Levine and Pedersen (2016) show that time-series momentum, moving-average crossover, and price-minus-moving-average rules are equivalent representations of a common filter.

2.2 Practitioner trend-following and breadth-momentum tactical allocation

Faber (2007) is a foundational reference paper, where he introduced a 10-month simple moving average rule applied to a 5-asset-class universe of US equities, foreign equities, commodities, REITs, and US Treasuries. The strategy holds each asset only when its price exceeds its moving average and rotates to cash otherwise. It achieves equity-like returns with significantly reduced drawdowns compared to buy-and-hold approaches. These findings have been replicated and extended in subsequent work (Faber 2013, 2017; Clare, Seaton, Smith, and Thomas 2016). Antonacci (2014) formalized the dual-momentum framework, which combines absolute and relative momentum. Known as the Global Equities Momentum (GEM) strategy, this was applied to a 3-asset-class universe of US equities, foreign equities, and US aggregate bonds.

Keller and Keuning (2016, 2017, 2018, 2023) and Keller (2022) developed a series of breadth-momentum strategies (PAA, VAA, DAA, BAA, and HAA). These were built around the concept of breadth momentum, which is defined as the cross-sectional count of universe constituents with positive momentum. Unlike the Faber-Antonacci approach, which evaluates each asset based on its own absolute momentum, breadth momentum evaluates the entire portfolio by counting the number of underperforming assets in the universe.

2.3 Hidden Markov Model regime detection

Regime-switching models for asset allocation was first introduced by Hamilton (1989). Ang and Bekaert (2002) applied the regime-switching framework to international equity allocation. Guidolin and Timmermann (2007) showed that regime-switching significantly improves allocation decisions across stocks, bonds, and bills. Bulla et al. (2011) applied Hidden Markov Models (HMM) specifically to tactical equity allocation. Nystrup, Madsen, and Lindström (2015, 2017, 2018, 2020) refined HMM specifications for TAA. They focused on walk-forward training, persistent-state penalization to reduce regime-flip turnover, and dynamic allocation under regime shifts.

A recent subset of the literature combines HMMs with active allocation rules. Kim et al. (2019) apply HMMs to time individual assets. Wang et al. (2020) use HMMs to switch between factor models on the S&P 500. Gupta et al. (2025) ensemble HMMs with tree-based classifiers.

2.4 Country-level equity momentum

Asness, Liew, and Stevens (1997) show that characteristics predicting the U.S. equity cross-section also predict the cross-section of international equity indices. Subsequent studies by Bhojraj and Swaminathan (2006), Chan, Hameed, and Tong (2000), and Rouwenhorst (1998) document robust 3- to 12-month momentum across international equity indices, attributing these effects primarily to within index predictability rather than currency movements. Andreu, Swinkels, and Tjong-A-Tjoe (2013) demonstrate that country momentum gross returns persist after accounting for realistic ETF transaction costs, which confirms the tradability of the universe used in this paper. Zaremba (2019) provides a comprehensive review of cross-sectional country equity returns.

3. Data and Universe

3.1 Universe

The risky universe consists of 19 country equity exchange-traded funds (ETFs) that cover the major developed and emerging equity markets. The defensive leg is a rotation between the iShares 20+ Year Treasury Bond ETF (TLT) and short-duration cash, where cash returns are given by BIL after May 2007 and by FRED 3-month T-bill returns prior to that date. The ticker, country and inception date of the assets in the risky and defensive universe is presented in Table 1. At each month-end date, the defensive position is set to TLT when its trailing 3-month total return is positive, or BIL when its trailing 3-month total return is negative.

Table 1. Asset universe.

Ticker	Country	Inception
SPY	United States	January 1993
EWC	Canada	March 1996
EWU	United Kingdom	March 1996
EWG	Germany	March 1996
EWQ	France	March 1996
EWI	Italy	March 1996
EWP	Spain	March 1996
EWN	Netherlands	March 1996
EWD	Sweden	March 1996
EWL	Switzerland	March 1996
EWJ	Japan	March 1996
EWA	Australia	March 1996
EWH	Hong Kong	March 1996
EWS	Singapore	March 1996
EWZ	Brazil	July 2000
EWX	Mexico	March 1996
EWY	South Korea	May 2000
EWT	Taiwan	June 2000
EZA	South Africa	February 2003
TLT	20+ year US Treasuries	July 2002
BIL	Cash (1 to 3 months US Treasuries)	May 2007

The composition of the universe follows the convention of the country-momentum literature. Asness, Liew, and Stevens (1997) used 18 MSCI country indices, Bhojraj and Swaminathan (2006) used approximately 38 country indices, and Andreu, Swinkels, and Tjong-A-Tjoe (2013)

used individual country ETFs to test if it is possible to trade the underlying country indexes. The universe includes individual country ETFs and excludes broad regional aggregates (i.e. EFA, EEM, VGK) and US sector ETFs (i.e. QQQ).

The binding inception constraint is that of EZA (inception February 2003). All 19 country equity ETFs have at least 6 months of trailing return data by August 2003. The backtesting start date is January 2004. The ETF prices are obtained from Yahoo Finance in adjusted close form (dividends reinvested, splits adjusted). The defensive leg uses TLT (inception July 2002) and BIL (inception May 2007), with FRED 3-month T-bill rates substituted before BIL inception. The HMM rate features are constructed from Yahoo Finance $\wedge$ TNX (10-year Treasury yield) and $\wedge$ IRX (3-month Treasury yield) series. Both rate series are available from before the start of the backtesting period.

3.2 Data conventions

Returns are computed using total returns, with dividends reinvested included in the adjusted close prices. The data is obtained from Yahoo Finance. Monthly observations are taken at the close of the last trading day of each calendar month. The positions are rebalanced on the open of the first trading day of the following month. This convention follows Faber (2007) and Keller and Keuning (2017), where signals are computed at month-end, execution occurs at the next available open, and there is no intra-month rebalancing.

3.3 Trade execution conventions

All buy and sell prices are recorded at the open of the first trading day of the month following the signal date. This approach avoids look-ahead bias because the signal is computed at time t and the trade is executed at time $t+1$. Baseline performance tables assume zero transaction costs to

maintain consistency with the published TAA literature. Results adjusted for transaction costs of 3, 5, and 10 basis points per round-trip switch are provided as a robustness check in Section 6.

The defensive position, consisting of either TLT or BIL, is held when the composite regime signal indicates a risk-off environment. In all other scenarios, the strategy allocates equally to the top 3 momentum ETFs.

3.4 Benchmarks

Performance is benchmarked against 5 reference strategies. The equal-weighted basket of the 19 equity country ETFs serves as the primary passive benchmark. EFA (MSCI EAFE) functions as a secondary passive benchmark. The SPY buy-and-hold is included as a reference benchmark for completeness, although it reflects a distinct investment universe of US large-cap equities rather than global country equities. The breadth-only and HMM-only strategies apply the same allocation rule using each single signal in isolation, so that we can isolate the marginal contribution of each regime detector.

4. Methodology

4.1 Breadth signal

At each month-end t , the breadth signal is computed as the fraction of the 19 risky ETFs with a positive trailing 6-month total return:

$$breadth(t) = \frac{1}{19} \sum_i 1\{r_i^{(6)}(t) > 0\}, \quad (1)$$

where $r_i^{(6)}(t)$ is the trailing 6-month total return of ETF i at month-end t , and the indicator function is 1 if the trailing return is positive and 0 otherwise.

The breadth signal ranges from 0, indicating no ETFs with a positive 6-month return, to 1, where all 19 ETFs are positive. The breadth gate flips to bear when $\text{breadth}(t) < 0.5$ and remains bullish otherwise. The 0.5 threshold corresponds to the median state of the universe. Robustness to the threshold choice is reported in Section 6.

4.2 Hidden Markov regime signal

We specify a three-state Hidden Markov Model (HMM) on four equity features of the equal-weighted 19 country ETFs and two rate-based features. The four equity features are the 1-day total return of the basket, the 21-day rolling realized volatility of the basket, the 63-day total return of the basket, and the 126-day rolling drawdown of the basket. The two rate features are the 21-day change in the 10-year Treasury yield ($\hat{\text{TNX}}$), which captures rate-momentum regimes, and the yield-curve slope ($\hat{\text{TNX}} - \hat{\text{IRX}}$), which is the 10-year minus 3-month yield differential and captures monetary-policy regimes including curve inversions.

These features summarize three dimensions of regime state, including instantaneous direction, volatility regime, and drawdown state, following the convention of the HMM-TAA literature (Nystrup et al. 2017, 2018; Bulla et al. 2011). The features are computed daily, regime inference is performed daily, and the posterior probability used in the strategy is the value at the month-end close of t .

An HMM specifies a discrete latent state sequence $Z_1, Z_2, \dots, Z_T$ with $Z_t \in \{1, \dots, K\}$ and an observed sequence of feature vectors $x_1, x_2, \dots, x_T$ with $x_t \in \mathbb{R}^d$. In our specification $K = 3$ and

$d = 6$. The model is parameterized by an initial-state distribution $\pi = (\pi_1, \dots, \pi_K)$ with $\pi_k = P(Z_1 = k)$, a $K \times K$ row-stochastic transition matrix A with entries $A_{ij} = P(Z_t = j | Z_{t-1} = i)$ satisfying $\sum_j A_{ij} = 1$, and a state-conditional emission density $b_k(x_t) = p(x_t | Z_t = k)$.

Following the convention of the HMM-TAA literature, we adopt a multivariate Gaussian emission,

$$x_t | Z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad (2)$$

where $\mu_k \in \mathbb{R}^d$ is a state-specific mean vector and $\Sigma_k \in \mathbb{R}^{d \times d}$ is a positive-definite full covariance matrix.

The full parameter set is $\theta = \{\pi, A, \{\mu_k, \Sigma_k\}\}$. The HMM imposes two conditional-independence assumptions. The hidden chain is first-order Markov, so the future state depends only on the present state. Observations are conditionally independent given the latent state, so the basket-level features carry no information beyond the current regime. The joint likelihood of a complete observation and state sequence is:

$$p(x_{1:T}, Z_{1:T} | \theta) = \pi_{Z_1} \cdot b_{Z_1}(x_1) \cdot \prod_{t=2}^T A_{Z_{t-1}, Z_t} \cdot b_{Z_t}(x_t), \quad (3)$$

Posterior state probabilities are computed by the forward-backward recursions.

The forward variable $\alpha_t(k) = p(x_{1:t}, Z_t = k | \theta)$ accumulates evidence from past observations and is initialized as $\alpha_1(k) = \pi_k \cdot b_k(x_1)$, and propagated by

$$\alpha_t(k) = \left[\sum_j \alpha_{t-1}(j) \cdot A_{jk} \right] \cdot b_k(x_t), \quad (4)$$

The backward variable $\beta_t(k) = p(x_{t+1:T}|Z_t = k, \theta)$ accumulates evidence from future observations, initialized as $\beta_T(k) = 1$, and propagated backward by

$$\beta_t(k) = \sum_j A_{kj} \cdot b_j(x_{t+1}) \cdot \beta_{t+1}(j), \quad (5)$$

The marginal observation likelihood is $p(x_{1:T}|\theta) = \sum_k \alpha_T(k)$, and the smoothed posterior over states uses information from the entire sequence,

$$\gamma_t(k) = P(Z_t = k|x_{1:T}, \theta) = \alpha_t(k) \cdot \beta_t(k) / \sum_j \alpha_t(j) \cdot \beta_t(j), \quad (6)$$

Smoothed posteriors are unavailable for real-time trading because future observations are not known at time t . We use the filtered posterior, which conditions only on observations available through t ,

$$P(Z_t = k|x_{1:t}, \theta) = \alpha_t(k) / \sum_j \alpha_t(j), \quad (7)$$

Model parameters are estimated by maximum likelihood. The observed-data log-likelihood is given by

$$\ell(\theta) = \log p(x_{1:T}|\theta), \quad (8)$$

$\ell(\theta)$ is non-convex in θ , and is optimized by the Baum-Welch algorithm. In the E-step, the forward-backward recursions yield the state posterior $\gamma_t(k)$ and the pairwise state posterior is given by

$$\xi_t(i, j) = P(Z_{t-1} = i, Z_t = j|x_{1:T}, \theta) = \alpha_{t-1}(i) \cdot A_{ij} \cdot b_j(x_t) \cdot \beta_t(j) / p(x_{1:T}|\theta), \quad (9)$$

In the M-step, parameters are updated by maximizing the expected complete-data log-likelihood.

The updates are available in closed form:

$$\pi_k \leftarrow \gamma_1(k), \quad (10)$$

$$A_{ij} \leftarrow \left(\sum_{t=2}^T \xi_t(i, j) \right) / \left(\sum_{t=2}^T \sum_{j'} \xi_t(i, j') \right), \quad (11)$$

$$\mu_k \leftarrow \left(\sum_t \gamma_t(k) x_t \right) / \left(\sum_t \gamma_t(k) \right), \quad (12)$$

$$\Sigma_k \leftarrow \left(\sum_t \gamma_t(k) (x_t - \mu_k)(x_t - \mu_k)^T \right) / \left(\sum_t \gamma_t(k) \right), \quad (13)$$

The E and M steps are iterated until the change in the observed-data log-likelihood falls below the hmmlearn default tolerance or a maximum iteration count is reached. Baum-Welch converges to a local maximum of the likelihood surface, and the solution depends on initialization.

The HMM is trained walk-forward on a five-year rolling window and refit every 12 months. Training is performed using the Baum-Welch algorithm. The risk-off state is identified as the one with the highest mean realized volatility and lowest mean return in the training window. This state identification is performed once per refit and held constant for the subsequent 12 months of out-of-sample inference.

The HMM gate flips to bear when the posterior probability of being in the risk-off state is at least 0.5, and remains bullish otherwise. The 0.5 threshold is the natural decision boundary for a binary classifier. Robustness to the number of HMM states (2, 3, 4), the posterior threshold (0.3 to 0.7), the training window length (3, 5, 7 years), and the refit frequency (6, 12, 24 months) is reported in

Section 6. The feature set is fixed ex ante based on prior literature precedent and is not optimized over.

4.3 Composite regime signal

The composite regime signal combines the breadth and HMM gates using an AND rule. $is_bear(t)$ is true only when both $breadth_bear(t)$ and $hmm_bear(t)$ are true. The strategy goes defensive only when both gates indicate risk-off. Equity allocation is held if at least one gate is bullish.

We chose to combine both signals instead of using them independently. The breadth signal is a cross-sectional regime estimator that measures how many country markets in the universe are in a positive trend regime. High breadth shows gains are broadly distributed across countries, while low breadth indicates narrow participation and that most countries have lost their positive 6-month trend.

The HMM signal is a time-series regime estimator. It infers the latent state of the basket from its returns, volatility, and drawdown dynamics, without reference to the cross-sectional composition of the universe. A high HMM risk-off posterior indicates that the recent dynamics of the basket resemble those of bear regimes in the training data.

These signals provide distinct information and may diverge. The universe may exhibit low breadth (cross-sectionally narrow), while the basket's time-series dynamics remain calm. Conversely, the basket may display high realized volatility and a sharp drawdown (high HMM risk-off posterior) even as breadth remains strong. In each case, one gate produces a bear flag that the other does not confirm.

Each signal taken alone produces a substantial number of false-positive bear flags. Breadth flips defensive at narrow-leadership tops that do not necessarily lead to sustained drawdowns, and remains defensive during extended post-capitulation recoveries. The HMM shifts to its high-volatility state during transient volatility spikes that do not result in a regime change. In both cases, there is a cost of spending time in defensive signals that turn out to be benign. This causes a drag on cumulative return that is not compensated by the drawdown control benefit.

The AND rule addresses this issue. We only go defensive when both the cross-sectional and time-series signals agree. This approach filters out single signal false positives and triggers only when both sources of evidence indicate a regime transition. These correspond to the major crises of the backtested period, including the 2008 financial crisis, the 2020 COVID drawdown, and the 2022 rate-driven bear. The alternative rule OR that flips defensive whenever either signal indicates risk-off tends to over-defend. Robustness to the combination rule is reported in Section 6.

4.4 Allocation rule

At each month-end t , the strategy executes the following sequence of steps:

1. Compute the trailing 6-month total return $r_t^{(6)}(t)$ using Eq. (1) for each of the 19 risky ETFs.
2. Compute the breadth signal and its associated bear gate $\text{breadth_gate}(t)$.
3. Compute the HMM posterior $P(\text{risk_off} \mid \text{features}_t)$ as described in Section 4.2, and set the HMM gate $\text{hmm_bear}(t)$.
4. Form the composite gate $\text{is_bear}(t) = \text{breadth_bear}(t) \text{ AND } \text{hmm_bear}(t)$.

5. If `is_bear(t)` is FALSE, the strategy holds an equal-weighted basket of the 3 ETFs with the highest trailing 6-month total returns for the next calendar month.
6. If `is_bear(t)` is TRUE, the strategy holds the defensive asset for the next calendar month. The defensive asset follows a separate rotation rule. TLT is selected if its trailing 3-month return at the signal date is positive. Otherwise, BIL is chosen.

Trades are executed at the open of the first trading day of month $t+1$. The position is held until the next month-end, when Steps 1–6 are repeated.

5. Results

5.1 Baseline performance

The baseline performance of the composite strategy, the passive benchmarks and the two signal strategies are shown in Table 2. The results shown are for the backtested period from January 2004 to December 2025. CAGR is the geometric mean annualized return. Sharpe is the excess return over the risk-free rate divided by the standard deviation of monthly returns. Maximum drawdown is computed on monthly total returns. Calmar is CAGR divided by the absolute maximum drawdown. Time in defensive is the fraction of months in which the strategy held TLT rather than the equity sleeve.

Table 2. Performance comparison of the 6 strategies.

Strategy	CAGR	Volatility	Sharpe	Max DD	Calmar	Time in Defensive
Composite (AND)	12.15%	18.39%	0.72	-23.14%	0.52	12.88%
Breadth-only	11.40%	16.91%	0.72	-22.03%	0.52	31.44%
HMM-only	11.22%	17.47%	0.70	-23.14%	0.48	20.08%
EFA (MSCI EAFE)	6.42%	16.55%	0.46	-57.37%	0.11	-
Equal-weighted basket	8.30%	18.38%	0.53	-57.95%	0.14	-
SPY buy-and-hold	10.65%	14.54%	0.77	-50.78%	0.21	-

The composite strategy has a Sharpe ratio of 0.72 and a maximum drawdown of -23.1% , compared with 0.53 and -58.0% for the equal-weighted basket. Against EFA, the composite delivers 0.72 versus 0.46 in Sharpe, and -23.1% versus -57.4% in maximum drawdown. Against SPY buy-and-hold, the composite exceeds SPY's CAGR by 1.5% while more than halving the maximum drawdown (-23.1% versus -50.8%). However, the composite delivers 0.72 versus 0.77 in Sharpe, which is attributable to the 14.5% volatility of US large-cap relative to the 18.4% volatility of the country momentum concentration.

The breadth-only strategy delivers 0.72 Sharpe and -22.0% maximum drawdown, and the HMM-only strategy delivers 0.70 and -23.1% . The composite has better performance compared to the single signal strategies in CAGR, and has comparable Sharpe and maximum drawdown. The composite holds the defensive leg in 12.9% of the months, which is a much lower time-in-defensive compared to 31% for breadth-only and 20% for HMM-only.

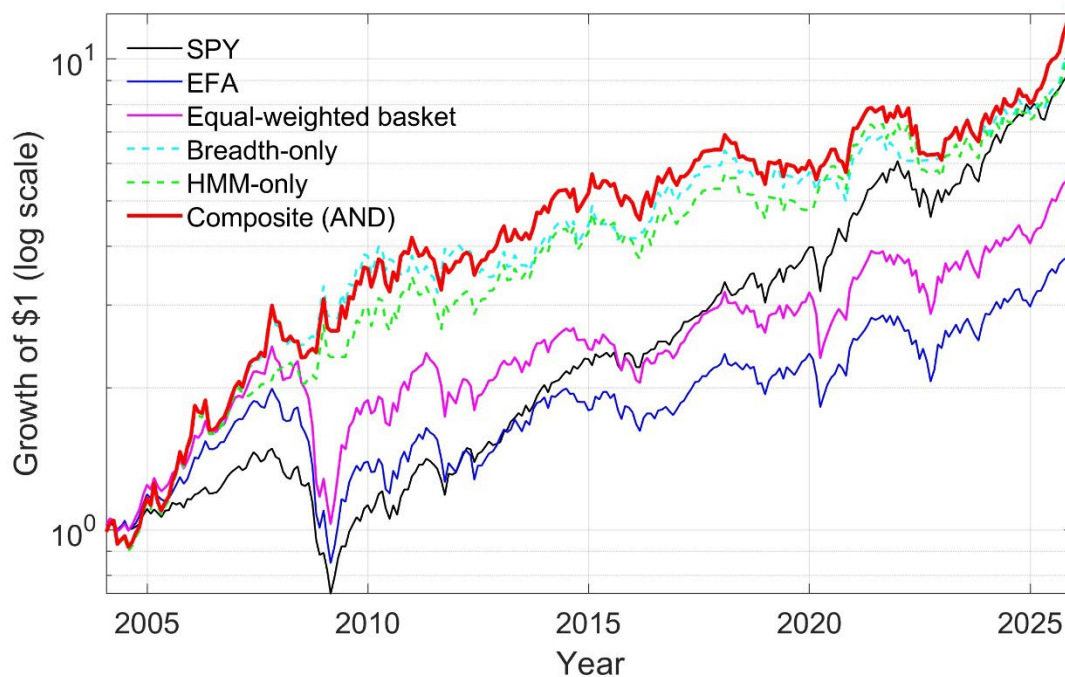


Figure 1. Equity curves for 6 strategies over the backtested period.

The equity curves for 6 strategies over the backtested period are shown in Figure 1. The composite strategy is compared with the three passive benchmarks (SPY, EFA, equal-weighted basket) and the two single-signal strategies (breadth-only, HMM-only). The composite strategy has the highest total returns, while EFA has the lowest total returns. The total returns for breadth-only and HMM-only strategies are comparable.

5.2 Per-year breakdown

The annual returns, maximum drawdown, time in cash, and number of position switches for both the composite strategy and the EFA (MSCI EAFE) benchmark are shown in Table 3.

Table 3. Annual performance breakdown.

Year	Composite	EFA (MSCI EAFE)	Excess	Composite Max DD	Time in Cash	Switches
2004	16.60%	18.96%	-2.36%	-12.20%	0.00%	10
2005	35.46%	13.32%	22.14%	-12.54%	0.00%	10
2006	27.11%	25.81%	1.30%	-13.59%	0.00%	11
2007	37.24%	9.95%	27.29%	-8.05%	0.00%	8
2008	12.24%	-41.03%	53.28%	-9.89%	83.33%	3
2009	15.90%	26.99%	-11.09%	-5.60%	33.33%	8
2010	16.47%	8.17%	8.30%	-14.71%	0.00%	11
2011	-11.08%	-12.23%	1.15%	-18.58%	33.33%	8
2012	14.76%	18.76%	-4.00%	-10.49%	8.33%	11
2013	14.11%	21.44%	-7.34%	-7.06%	0.00%	11
2014	6.73%	-6.19%	12.93%	-10.28%	25.00%	6
2015	-4.77%	-0.99%	-3.78%	-14.27%	16.67%	10
2016	9.28%	1.37%	7.91%	-7.66%	0.00%	11
2017	18.81%	25.08%	-6.26%	-0.03%	0.00%	11
2018	-15.49%	-13.82%	-1.67%	-21.43%	16.67%	10
2019	12.09%	22.04%	-9.95%	-7.55%	8.33%	11
2020	17.24%	7.60%	9.65%	-10.52%	33.33%	9
2021	11.38%	11.45%	-0.07%	-6.98%	0.00%	11
2022	-23.14%	-14.39%	-8.76%	-22.47%	25.00%	8
2023	25.21%	18.36%	6.84%	-9.87%	0.00%	8
2024	5.44%	3.49%	1.95%	-6.61%	0.00%	10
2025	54.66%	32.11%	22.55%	-1.62%	0.00%	11

The composite strategy outperforms EFA in 12 of the 22 years. The best performance is obtained when momentum concentration in specific country markets is high or when the regime gate flips defensive ahead of a global equity drawdown. The largest excess returns were observed during the 2008 financial crisis. The composite underperforms EFA in years of broad-based international rallies, such as 2009, 2013 and 2019.

The 2022 rate-driven bear is the largest single drawdown contributor, with the composite returning -23.1% compared to EFA's -14.4% . The rate-feature HMM activated in August 2022 and the defensive leg held cash through the second half of the year, which limited further damage.

The composite has approximately 9 position changes per year on average. The majority are equity-to-equity rotations within the top 3 basket, and equity-to-defensive flips are concentrated in the 2008, 2011, 2018, and 2020 transition periods.

5.3 Decomposition: cross-sectional and time-series regime contributions

The baseline result shows that the composite signal outperforms the single signal strategy. This section examines whether the breadth and HMM signals contribute non-redundant information.

We classify each month in the backtested period by the joint state of the two signals. The time spent in each cell of the two-by-two state matrix, and the mean forward return for each cell is shown in Table 4. Mean forward returns are the realized return of the strategy's equity position in the month following the signal date.

Table 4. Joint regime state classification and forward returns.

	breadth_bear = FALSE	breadth_bear = TRUE
hmm_bear = FALSE	Both bullish: 61.2% of months Mean forward return: 1.11%	Breadth-only bear: 18.6% of months Mean forward return: 0.75%
hmm_bear = TRUE	HMM-only bear: 7.2% of months Mean forward return: 1.74%	Both bearish: 12.9% of months Mean forward return: -1.36%

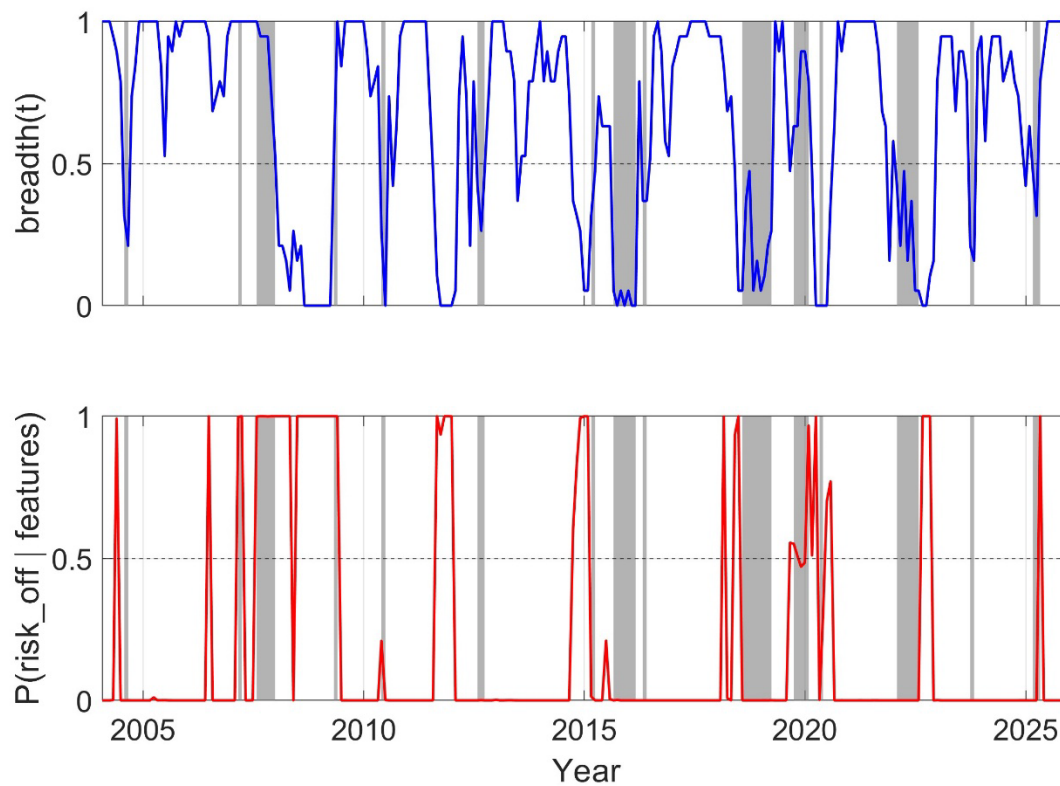


Figure 2. Time series of $\text{breadth}(t)$ on the upper panel and $P(\text{risk_off} \mid \text{features})$ on the lower panel.

The shaded vertical bands indicate periods in which the two signals disagree.

If the two signals were redundant, the off-diagonal cells, where one signal is bear and the other is not, would be infrequent. However, the results show the opposite. The off-diagonal cells together account for 25.9% of the months, and their mean forward equity returns are positive. Only the cell where both signals are bear exhibits a negative mean forward return of -1.36% . Each signal individually produces frequent false positives. The AND rule value is used to filter these false positives, so that the composite strategy goes to the risk-off state only when both signals agree. The periods in which the two signals agree (unshaded) and disagree (shaded) are shown in Figure 2.

5.4 Drawdown analysis

The worst drawdowns are concentrated at regime transition points and during gradual deteriorations, where the AND gate fires only after the equity drawdown has occurred. The drawdown of the composite strategy and SPY over the backtested period are shown in Figure 3. The composite strategy has the largest drawdown during the 2022 rate-driven bear, where the trough is -23.1% . The second largest drawdown for the composite of -22.6% occurred during the 2011 EU debt crisis.

The composite strategy has better risk-adjusted performance than passive multi-country benchmarks, as it avoids the deepest drawdowns during bear regimes and concentrate in the highest-momentum countries during bull regimes. The maximum drawdown for SPY is -50.8% , which is more than twice the composite strategy's maximum drawdown. Both the equal-weighted basket and EFA have a maximum drawdown of more than -57% . The composite strategy uses the AND gate and the rotating defensive leg can switch from TLT to cash when long-duration

Treasuries themselves are in drawdown. This provides a structural protection against rate-driven episodes (2022) that an unconditional TLT defensive lacks.

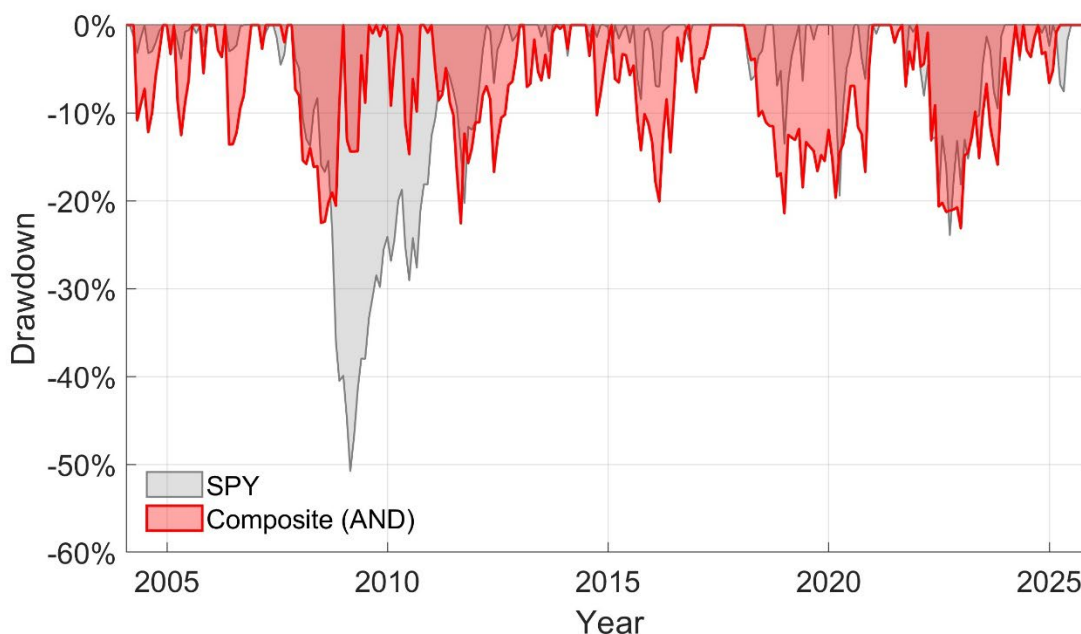


Figure 3. Drawdown of composite strategy and SPY over the backtested period.

6. Robustness

In this section, we evaluate the robustness of the baseline results to parameter choices. We vary one parameter at a time, while holding all other parameters at their baseline values. The baseline values are breadth threshold 0.5, HMM posterior threshold 0.5, three HMM states, five-year training window, 12-month refit frequency, 6-month momentum lookback, AND combination rule, top-3 equal-weighted equity sleeve, and TLT defensive leg.

6.1 Breadth threshold sensitivity

Table 5. Breadth threshold sensitivity.

Breadth threshold	CAGR	Sharpe	Max DD	Calmar	Time in Cash
0.30	11.34%	0.67	-29.80%	0.38	10.61%
0.40	11.15%	0.66	-31.59%	0.35	11.74%
0.50 (baseline)	12.15%	0.72	-23.14%	0.52	12.88%
0.60	12.07%	0.72	-23.14%	0.52	13.64%
0.70	11.92%	0.71	-23.14%	0.52	14.39%

The strategy's performance across various breadth threshold values is shown in Table 5. The baseline value is 0.50, which delivers the best performance of CAGR 12.15% and Sharpe 0.72. The performance degrades for threshold values both above and below 0.5. At higher thresholds the result is very similar to the baseline because the AND gate is constrained by the HMM signal.

6.2 HMM posterior threshold sensitivity

Table 6. HMM posterior threshold sensitivity.

HMM posterior threshold	CAGR	Sharpe	Max DD	Calmar	Time in Cash
0.30	12.15%	0.72	-23.14%	0.52	12.88%
0.40	12.15%	0.72	-23.14%	0.52	12.88%
0.50 (baseline)	12.15%	0.72	-23.14%	0.52	12.88%
0.60	11.28%	0.67	-29.74%	0.38	12.12%
0.70	11.22%	0.67	-29.74%	0.38	11.74%

The strategy's performance across a range of HMM posterior threshold values is shown in Table 6. At lower thresholds, the HMM gate is more sensitive and spends more time in cash. In contrast,

higher thresholds result in reduced sensitivity and spends less time in cash. A threshold of 0.50 is selected for the strategy, as it represents the natural decision boundary for a binary classifier. The Sharpe ratio is constant at 0.72 for HMM thresholds of 0.30, 0.40, and 0.50.

6.3 Number of HMM states

Table 7. Number of HMM states.

Number of states	CAGR	Sharpe	Max DD	Calmar	Time in Cash
2	12.28%	0.74	-33.00%	0.37	19.70%
3 (baseline)	12.15%	0.72	-23.14%	0.52	12.88%
4	9.18%	0.56	-37.65%	0.24	7.58%

The strategy's performance across varying numbers of states is shown in Table 7. A 2-state HMM models the basket with a single bullish and a single bearish regime. The baseline specification is a 3-state HMM which adds a transitional state. The 3-state model has a CAGR of 12.15%, Sharpe ratio of 0.72 and a maximum drawdown of -23.14%. The 2-state and 3-state HMM models perform comparably for CAGR and Sharpe ratio, although the 3-state model exhibits a smaller maximum drawdown. The 4-state model overfits the training window. A 4-state HMM has on the order of 100 parameters to estimate from a 5-year window, and out-of-sample classification tends to degrade.

6.4 HMM training window length and refit frequency

Table 8. HMM training window and refit frequency.

Training window	Refit frequency	CAGR	Sharpe	Max DD	Calmar
3 years	12 months	8.86%	0.55	-35.81%	0.25
5 years (baseline)	12 months	12.15%	0.72	-23.14%	0.52
7 years	12 months	11.84%	0.71	-31.53%	0.38
3 years	6 months	11.04%	0.66	-36.02%	0.31
5 years	6 months	13.74%	0.78	-22.58%	0.61
7 years	6 months	14.64%	0.88	-23.28%	0.63
3 years	24 months	9.43%	0.58	-35.81%	0.26
5 years	24 months	10.31%	0.63	-28.97%	0.36
7 years	24 months	11.26%	0.68	-35.01%	0.32

The strategy's performance across various training window and refit frequency values is shown in Table 8. Shorter training windows make the HMM more responsive to recent regime structure but more vulnerable to overfitting on small samples. In contrast, longer windows are more stable but slower to adapt to structural change. The baseline specification has a 5-year training window and a 12-month refit frequency.

The best performance is obtained with a 7-year training window and a 6-month refit frequency (CAGR 14.64%, Sharpe 0.88, Calmar 0.63). We found that more frequent (6-month) refitting is best at every window length, and the longer training windows (5-7 years) perform better than the 3-year window. There is a significant decline in performance for the 3-year window as there is too little data to fit a Gaussian HMM with features. We retain the 5-year training window and 12-

month refit frequency specification as the baseline as it was fixed ex ante rather than chosen from this grid.

6.5 Momentum lookback

Table 9. Momentum lookback sensitivity.

Momentum lookback	CAGR	Sharpe	Max DD	Calmar	Time in Cash
6 months (baseline)	12.15%	0.72	-23.14%	0.52	12.88%
9 months	8.05%	0.51	-49.41%	0.16	10.61%
12 months	8.92%	0.57	-36.24%	0.25	11.83%
15 months	8.24%	0.53	-37.77%	0.22	9.65%

The strategy's performance across various momentum lookback periods is shown in Table 9. The best performance was obtained with the 6-month momentum lookback, which is shorter than the conventional 12-month lookback used by Asness, Liew, and Stevens (1997) and others in the country-momentum literature.

6.6 Combination rule

Table 10. Combination rule sensitivity.

Combination rule	CAGR	Sharpe	Max DD	Calmar	Time in Cash
Breadth-only	11.40%	0.72	-22.03%	0.52	31.44%
HMM-only	11.22%	0.70	-23.14%	0.48	20.08%
OR	10.48%	0.71	-20.36%	0.51	38.64%
AND (baseline)	12.15%	0.72	-23.14%	0.52	12.88%
Average	10.49%	0.66	-23.14%	0.45	18.94%

The strategy's performance across various combination rules is shown in Table 10. The baseline specification is the AND rule, which requires both signals to indicate risk-off before flipping defensive. The OR rule flips defensive whenever either signal indicates risk-off. The average rule combines the two signals as a weighted probability of 0.5.

The AND rule produces the best performance with a CAGR of 12.15% and a Sharpe ratio of 0.72. The OR rule is a more defensive strategy with 38.64% time in defensive and a smaller maximum drawdown of -20.36% , but with a lower CAGR of 10.48%. The single-signal rules, using either breadth-only or HMM-only, deliver slightly lower CAGR of 11.40% and 11.22%, with Sharpe ratios of 0.72 and 0.70, respectively. The average rule has the worst performance with a CAGR of 10.49% and a Sharpe ratio of 0.66.

6.7 Cost sensitivity

Table 11. Cost sensitivity.

Cost per switch	CAGR	Sharpe	Max DD	Calmar
0 bps (baseline)	12.15%	0.72	-23.14%	0.52
3 bps	11.80%	0.70	-23.35%	0.51
5 bps	11.58%	0.69	-23.49%	0.49
10 bps	11.01%	0.66	-23.87%	0.46

The strategy's performance for a range of cost per switch values is shown in Table 11. Execution costs for ETFs of this liquidity profile range from 3 to 5 bps per round-trip switch. At 3, 5 and 10 bps, the strategy delivers 11.80% CAGR and 0.70 Sharpe, 11.58% CAGR and 0.69 Sharpe, and 11.01% CAGR and 0.66 Sharpe, respectively. The strategy averages approximately 9 position changes per year, which gives an annual cost drag of roughly 27–45 bps.

6.8 Summary

The baseline result shown in Section 5 is directionally robust to the parameter choices in the strategy specification. We conduct a grid search across 5,400 parameter combinations, spanning breadth thresholds $\{0.30, 0.40, 0.50, 0.60, 0.70\}$, HMM posterior thresholds $\{0.30, 0.40, 0.50, 0.60, 0.70\}$, HMM state counts $\{2, 3, 4\}$, HMM training-window lengths $\{3, 5, 7\}$ years, refit frequencies $\{6, 12, 24\}$ months, momentum lookbacks $\{6, 9, 12, 15\}$ months, and combination rules $\{\text{AND}, \text{OR}\}$. Every configuration produces a positive CAGR, and risk-adjusted performance is concentrated in the longer-window, frequent-refit, short-lookback region. The best configuration achieves a Sharpe ratio of 0.89, CAGR 14.56% and maximum drawdown -22.58%. Its specifications are a 3-state HMM, a 7-year training window, a 6-month refit frequency, a 6-month momentum lookback, a breadth threshold of 0.60, an HMM posterior threshold of 0.30, and the AND combination rule. The baseline specification (3-state HMM, 5-year window, 12-month refit, 0.50 breadth threshold) attains a Sharpe ratio of 0.72, which places it in the 96th percentile of the grid. The baseline was specified ex ante, and in the top tier of the parameter grid indicating that its performance is not an artifact of in-sample optimization. We therefore retain the more conservative baseline rather than the grid-optimal configuration, to align with prior literature and to avoid overfitting the specification.

7. Discussion

7.1 Why the AND rule works

Our results show that cross-sectional breadth and time series HMM must be combined for effective regime identification. When considered independently, each regime signal produces false positives. Breadth flips defensive at narrow leadership tops, which do not always precede sustained

drawdowns, and remains defensive during recovery periods. HMM flips to its high volatility state during transient shocks. Using the defensive leg on either signal alone results in 20-40% of the time spent in a defensive leg, leading to a corresponding drag on cumulative return.

This is shown by the decomposition in Section 5.3. The number of months in the off-diagonal cells of the signal state matrix account for more than 20% of the backtested period. Forward returns in these single signal bear months are not negative, which is consistent with the false-positive interpretation. Forward returns in the both bear cells are sharply negative, and these correspond to the high-conviction regime transitions on which the AND rule fires. The robustness analysis in Section 6 shows that the AND rule outperforms the OR rule. The OR rule spends nearly 40% of months in the defensive leg which leads to a corresponding drag on CAGR.

7.2 Limitations

We note that there are several limitations of the analysis.

Monthly evaluation frequency. The strategy evaluates at month-end and rebalances on the first trading day of the following month. This convention follows the practitioner TAA literature and supports clean implementation. However, intra-month regime breaks are not acted upon until the subsequent month-end evaluation. In sharp drawdowns that develop over the course of a single calendar month, such as the COVID-19 drawdown of February-March 2020, a strategy with daily evaluation could have flipped defensive earlier than the monthly version examined here.

Equity-sleeve concentration. The strategy holds an equal-weighted basket of 3 equity ETFs when bullish, and this diversifies away a portion of the idiosyncratic country risk that a top-1 holding would carry. The choice of $N = 3$ is conventional in the breadth-momentum literature.

Backtested period and bear regime count. The backtested period spans 21 years and contains a limited number of bear regimes. Statistical power on bear regime performance is limited. The performance in future bear regime depends on how similar it is to the bear regimes in the backtested period.

HMM specification stability. The HMM is specified with 4 basket-level features (the 1-day return, the 21-day realized volatility, the 63-day return, and the 126-day drawdown). We do not optimize over the feature set, and alternative feature constructions might have different strategy behavior.

Cost assumptions. The baseline performance is reported gross of costs to maintain consistency with the published TAA literature. Cost-adjusted performance at 3, 5, and 10 basis points per switch is reported in Section 6.7. Costs in actual execution depend on order size, market conditions, and execution strategy, which is not captured in the backtest.

Asset universe construction. The asset universe consists of 19 country equity ETFs and excludes other asset classes. This construction follows the country momentum literature convention, but a multi-asset extension would be a natural follow-up.

7.3 Future work

In this study, we implement a basic regime conditional country momentum strategy. Several extensions are worth exploring.

Increasing the frequency of position rebalancing to daily or weekly could improve detection of intra-month regime changes. However, this would increase turnover and sensitivity to noise. These challenges may be overcome by persistence techniques.

Extending the asset universe to include bonds, commodities, or other asset classes would test whether the approach generalizes beyond single asset class equity rotation. A multi-asset universe would also enable comparison to the Faber, Antonacci and Keller TAA strategies using identical universe constructions.

Alternative time-series regime classifiers such as Markov-switching models, jump models, threshold autoregressive models, and machine-learning classifiers may be explored to enhance regime detection.

8. Conclusion

We examined the use of two regime detection methods, breadth momentum and Hidden Markov Models (HMM), in a country momentum tactical allocation strategy. The composite strategy is applied to a universe of 19 country equity ETFs.

The baseline result of the composite strategy, which combines both regime signals through an AND rule, shows a Sharpe ratio of 0.72 and a maximum drawdown of -23.1% . The composite strategy outperforms the equal-weighted basket, which has a Sharpe ratio of 0.53 and a maximum drawdown of -58.0% . It also surpasses the MSCI EAFE, with a Sharpe ratio of 0.46 and a maximum drawdown of -57.4% . The composite signal more than halves the maximum drawdown of the MSCI EAFE and exceeds SPY's CAGR by 1.5% (12.2% versus 10.7%).

The reason for combining both signals is that breadth momentum and HMM regime detection capture distinct information. Breadth measures the cross-sectional dispersion of the universe, while HMM measures the time-series statistical state of the basket. Each signal on its own can lead to false positives. The AND rule, which requires both signals to confirm before shifting to a defensive

stance, filters out these false positives. The signal indicates a bear market only when there is strong conviction. This corresponds to the major crises during the backtested period in 2008, 2020 and 2022.

The robustness analysis shows that the baseline result remains stable across various parameter choices. The results are consistent across a 5,400-combination grid that varies the breadth threshold, HMM threshold, number of HMM states, training-window length, refit frequency, momentum lookback, and combination rule. The high number of successful combinations across the parameter grid indicates that the strategy captures a genuine signal in the data rather than being a result of parameter tuning.

However, there are some limitations in the backtested results. The monthly evaluation frequency means intra-month regime changes are not acted upon quickly. ETF concentration during bull months exposes the portfolio to idiosyncratic country risks. The backtested period contains a limited number of bear regimes, which reduces statistical power regarding bear regime performance. The HMM feature specification is fixed ex-ante and not chosen through optimization. Future work may include improvements such as more frequent evaluations, top-N concentration, multi-asset universes, alternative regime detection methods, and probabilistic combination rules.

About the Author

Vivien Pei Wen Chua is Chief Investment Officer and co-founder at Meadowfield Capital, and founder of One Degree Research. She began her career as an Assistant Professor at the National University of Singapore, where she led research at the intersection of engineering and applied quantitative methods. She holds a Ph.D. and M.S. in Engineering from Stanford University and a B.S. in Engineering from the Georgia Institute of Technology. Her work has been recognised internationally. Vivien was named to the MIT Technology Review Innovators Under 35 (Asia Pacific) list. She has also been honoured as one of Singapore's 100 Women in Tech and named among Insurance Business Elite Women.

About Meadowfield Capital

Meadowfield Capital is an independent fund management company, regulated by the Monetary Authority of Singapore under a Capital Markets Services License. The firm manages investments across asset classes with the objective of delivering superior risk-adjusted returns to its clients in support of long-term growth.

About One Degree Research

One Degree Research is the research lab of Meadowfield Capital. Its work focuses on quantitative finance, including regime detection across global markets, and the development of systematic global macro and multi-asset portfolios. The firm's intellectual foundations draw on two decades of peer-reviewed research in mathematical physics, computational fluid dynamics, and large-scale numerical simulation.

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